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Dear Professor Marc & Selection Team,

I am writing to apply for the EngD position on the **interactive reverse logistics map** for circular construction (CIRCOLOGIC). In construction we are already good at taking buildings apart, but a lot of good material still ends up as waste. From what I understand, the main problem is information. People often do not know which materials are available, in what condition, where they are, or when they will need them. So they choose new materials instead of reused ones. A map that shows what is available, a way to calculate cost and CO_2, and a simple way to compare options would make reuse a much easier choice. That is the kind of solution I would like to help build.

I hold a Master's in Environmental Engineering and a Bachelor's and Diploma in Civil Engineering. In my master's thesis, **Agile Project Management in the Construction Industry**, I studied how Agile can help construction teams work better together, manage risk, make decisions and use resources more carefully. Completing that work also opened a wider interest for me: what happens to materials when a building reaches the end of its life. From there I looked into circular construction and reverse logistics, and into earlier work such as Professor Voordijk's research on recovering building elements for reuse. His work shows that reuse needs coordination from the donor building to the target building, not only new technology. An interactive reverse logistics map is a step further than logistics and Circular Matching Platforms like DuSpot, especially when timing, quality and cost are uncertain.

At Aug.e I worked on digital twins for smart buildings. We used a digital model to understand and improve how a real building performs. At Hive CPQ I managed large product datasets from manufacturers and turned complex information into simple tools that non-technical users could work with. At iostring I worked with a construction partner on a circular construction hypothesis: how Agile project management can bring circular practices into real projects. From these roles I bring experience with construction data, stakeholders, digital tools and GIS. I have been on a career break to care for my newborns, and I kept learning about this field during that time.

In this EngD I want to learn reverse-logistics modelling in more depth, get better at using GIS to match donor and target materials, and learn how to build and test a decision-support map with industry partners. I would like to do this at the University of Twente, where the programme mixes advanced study with a real design project. My plan is to take the circular construction work further: start from real use cases, test the map in a (de)construction pilot with partners like Dura Vermeer and TNO, and improve it with the people who will use it. I can relocate to the Netherlands.

Thank you for reading my application. I would be glad to talk further.

Yours sincerely,
Pooja Awasare